Localize Any Object in X-Ray Security Scans without Human Annotation

Anonymous Author(s)
Affiliation
Address
email

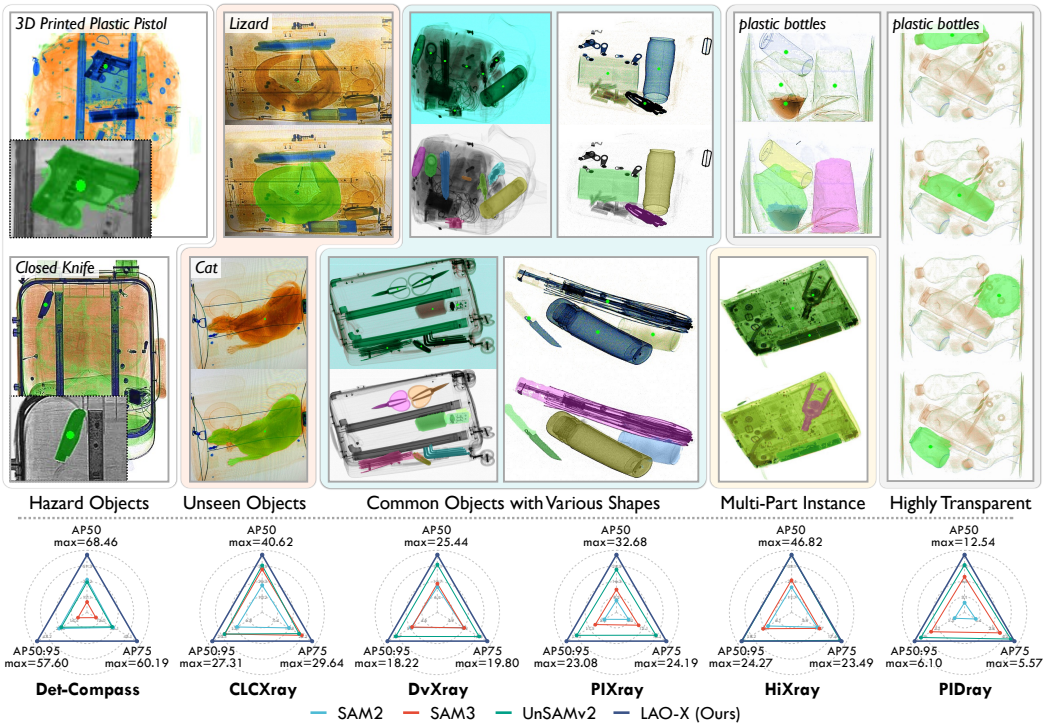


Figure 1: **Any-object localization in X-ray screening. Top:** Point-prompted segmentation on challenging X-ray cases, including *hazard objects*, *unseen objects*, *common objects with diverse shapes*, *multi-part instances*, and *highly transparent* items. The green dot indicates the point prompt. Different colors denote predicted masks. **Bottom:** Radar plots of **AP50**, **AP75**, and $\mathbf{AP}_{50}^{95}$ on six benchmarks (Det-Compass, CLCXray, DvXray, PIXray, HiXray, PIDray), comparing SAM2, SAM3, UnSAMv2, and **LAO-X (Ours)**. LAO-X consistently achieves the strongest performance.

Abstract

Universal object localization in X-ray security inspection is critical for automated threat detection in safety-critical venues. However, unlike everyday RGB images that dominate web-scale visual data, X-ray scans exhibit distinct color patterns, ambiguous boundaries, and compositional structures caused by volumetric superposition. These gaps hinder the direct zero-shot transfer of dense perception foundation models trained on web-scale RGB data. Moreover, annotated X-ray data is scarce and requires expert labeling, limiting both the training of generalizable

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

X-ray native models and the adaptation of RGB foundation models for X-ray data via fine-tuning. Given these challenges, the bright promise of highly generalizable perception models, enabled by data scaling laws in the RGB domain, remains largely out of reach for X-ray inspection. To this end, we introduce LAO-X, a fully self-supervised framework that Locates Any Object in X-ray scans using diverse synthesized image–annotation pairs with granularity-aware supervision. LAO-X first designs a saliency-guided X-ray object mining module to separate diverse object instances, which are then used for physics-guided synthesis in the absorbance domain. LAO-X further incorporates an occlusion-controlled curriculum strategy to fine-tune a Segment Anything Model 2 (SAM2) localizer, progressively adapting it to X-ray scans with increasing object counts and overlap levels. Experiments on six X-ray benchmarks show that LAO-X substantially improves category-agnostic localization, achieving 2% to 23% mAP gains over SAM2 and X-ray specific baselines in heavily cluttered scenarios, entirely without human-annotated labels.

1 Introduction

X-ray security screening is a critical safeguard for detecting concealed dangerous items in safety-sensitive scenarios, such as airports and large public events [37]. Although X-ray imaging enables non-intrusive inspection, it still relies on well-trained professionals to manually examine cluttered and overlapping contents. The consistent expansion of global mobility and logistics poses the demand for fast and reliable inspection systems, challenging operational *efficiency*. Meanwhile, advances in manufacturing technologies facilitate the creation of irregular and previously unseen threats (*e.g.*, 3D-printed weapons), further complicating *generalized* object detection [11].

Traditional automated X-ray screening methods typically rely on fully-supervised detectors [6, 20, 27] trained on pre-defined categories. However, the high cost of expert annotation limits the diversity of object categories and scanning scenes, making these models brittle to unknown objects and domain shifts across scanners. In contrast, RGB vision has entered the foundation-model era, with methods such as the Segment Anything Model (SAM) series [17, 32, 4] pre-trained at scale to learn transferable objectness priors and segment "*anything*" in a zero-shot manner. However, such RGB-based foundation models remain unreliable on X-ray imagery, where surface textures and visible boundaries are replaced by density cues, transparency, and projective superposition (Fig. 1).

The generalized localization capability of SAM-based foundation models is derived from diverse objects and comprehensive mask annotations in large-scale RGB datasets. In X-ray security screening, however, acquiring scans requires specialized equipment, and exhaustively annotating existing scans is impractical due to clutter, weak boundaries, severe overlap, and scanner-dependent appearance. As a result, existing X-ray datasets lack the category diversity and dense mask supervision required to train a generalized localizer from human annotations. At the same time, directly reusing RGB objectness priors is insufficient, since X-ray objects are not defined by visible surfaces alone but by projected density patterns under physical superposition. Facing these challenges, a central question arises: *Can we transfer the "localize any object" capability of RGB foundation models to X-ray security screening without human annotations?*

We answer this question with **LAO-X**, a self-supervised framework for Locating Any Object in X-ray scans. LAO-X adapts the base localizer SAM2 [32] to the X-ray modality using physics-guided pseudo samples generated and annotated entirely from unlabeled X-ray data. Specifically, LAO-X first mines salient foreground regions from unlabeled classification scans and extracts object-centric masks to construct an X-ray object bank. Then, we design physics-guided X-ray inspection synthesis to recompose the mined objects into controllable scenes in accordance with the Beer–Lambert law [38]. Since a synthesized X-ray object can provide both a reliable local component and a holistic instance extent, we organize the generated pseudo masks along a local-to-holistic granularity scale: holistic instance masks supervise object-level localization, while over-segmented local component masks provide complementary fine-structure supervision under weak boundaries and overlap. Finally, LAO-X trains a granularity-conditioned SAM2 [45] under an occlusion-controlled curriculum [9], progressively increasing scene complexity from sparse objects to severe superposition. This design converts unlabeled X-ray scans into structured, physics-aware supervision, enabling stable category-agnostic localization on real scans without manual labels. Extensive experiments on X-ray security screening datasets demonstrate the superiority of LAO-X. Our method achieves state-of-the-art

performance in both mAP and AR, surpassing baselines such as SAM3 [4] and domain-specific open-vocabulary detectors like RAXO [12] by up to **23.48%** points across objects from over 300 categories. Qualitatively, LAO-X produces cleaner boundaries under complex attenuation and recovers finer structures across multi-scale threats. Overall, our contributions are *threefold*:

- We address an underexplored foundation for X-ray security perception: localizing objects before recognizing them. Unlike closed-set or open-vocabulary detectors that rely on predefined or prompted semantics, our setting targets category-agnostic X-ray objectness without category names, text prompts, visual exemplars, or target-domain annotations.
- We propose an systematic X-ray synthesis-and-labeling pipeline that converts unlabeled scans into physics-consistent image-annotation pairs. It mines object-centric regions, re-composes them in the Beer-Lambert absorbance domain, and generates structured local-to-holistic pseudo masks without human labels.
- We evaluate LAO-X on six test-only X-ray benchmarks to assess cross-dataset generalization from unlabeled source scans. Without target-domain training data, LAO-X achieves stronger category-agnostic localization than promptable foundation models and self-supervised segmentation baselines, and also transfers effectively to threat-item benchmarks.

2 Related Work

Object Detection in X-Ray Security Screening. Conventional automated X-ray security inspection methods typically follow a fully supervised paradigm [6, 20, 27]. These methods operate under a closed-set assumption [40, 42, 35], which learn to localizing predefined categories from expert annotations. However, existing benchmarks such as CLCXray [46] and PIXray [25] contain fewer than 20 categories, largely due to the high cost of annotating complex X-ray scans with severe overlap. As a result, fully supervised methods often exhibit limited generalization when encountering out-of-distribution objects and scenes. To address evolving object distributions in luggage screening, recent works explore open-vocabulary detection with large-scale vision-language models such as CLIP [31]. OVXD [22] adapts CLIP to X-ray imagery using three learnable adapters that aggregate visual cues. RAXO [12] replaces the text-based classifiers of RGB detectors with X-ray visual class descriptors built from in-house and web-retrieved samples X-SSL [29] combines zero-shot proposals, multimodal clustering, and knowledge distillation for self-supervised threat localization. Despite covering broader categories, open-vocabulary methods still require category names or visual exemplars at inference time. In contrast, our LAO-X enables *category-agnostic object localization* in X-ray images by learning localization priors from physics-guided synthesized samples. *By removing the dependence on text prompts, web retrieval, and target-domain annotations, LAO-X shifts the focus from semantic recognition to physics-grounded localization under X-ray superposition*.

Foundation Models for Universal Localization. Promptable segmentation foundation models have reshaped generic object localization. The Segment Anything Model (SAM) [17] and its variants (SAM2 [32], SAM3 [4]) achieve strong zero-shot, class-agnostic segmentation by converting sparse prompts (*e.g.*, points) into masks, benefiting from large-scale training on SA-1B [17] (11M images, 1B masks). Despite strong generalization on natural images, SAM becomes unreliable under extreme object superposition, weak boundaries, and low-density materials [13]. A common solution involves domain adaptation via supervised fine-tuning or lightweight adapters [8, 7, 21], but these methods require substantial labeled data, which is impractical for X-ray imagery. Recent works explore multi-granularity segmentation [19, 47] and unsupervised signals for foundation models [45]. However, their assumptions remain largely rooted in RGB imagery, where object boundaries and part-whole structures are visually separable. *This highlights the need for localization priors that are not only category-agnostic, but also aware of the physics-induced structural ambiguities of X-ray imaging*.

Self-Supervised Segmentation and Pseudo-Labeling. Self-supervised visual features, especially those from ViTs, often exhibit object-aligned structures and have been widely used for label-free segmentation and proposal generation [5, 30, 36]. CutLER [43] leverages normalized cuts [34] to discover object masks, while SOHES [3], UnSAM [44], and UnSAMv2 [45] further learn multi-granularity predictions from hierarchical pseudo-labels generated through divide-and-conquer pipelines. These methods provide useful unsupervised signals, but they largely rely on RGB-style appearance grouping and visible-surface objectness. Such assumptions are poorly suited to X-ray scans, where objects are projected through nonlinear attenuation and multiple items may occupy

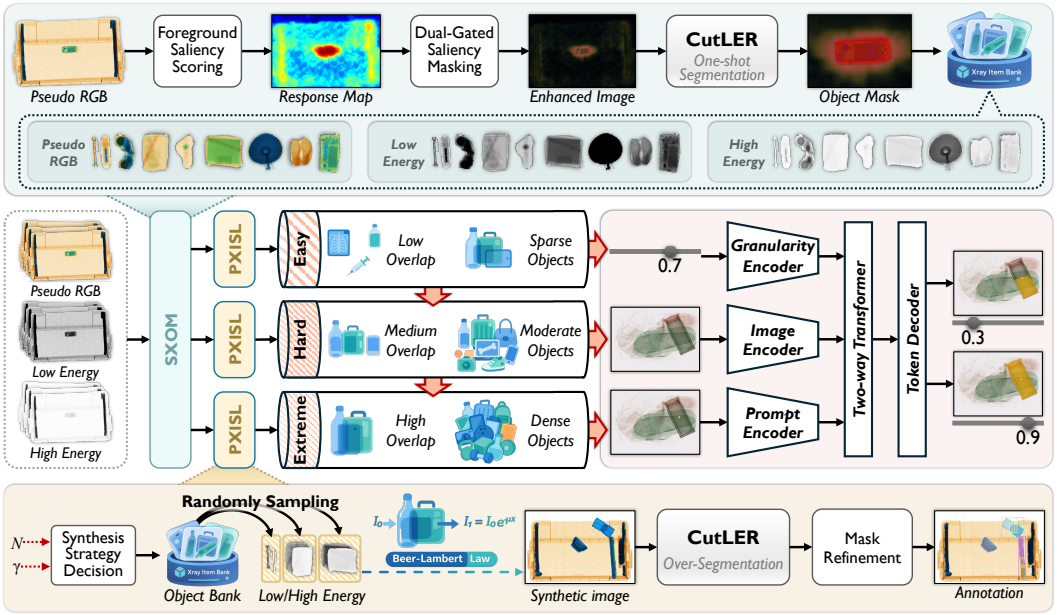


Figure 2: **Overview of LAO-X**. Giving an unlabeled X-ray scans, we first extracting foreground objects to build an item bank. Then synthesize and labeling images with randomly sampled objects to simulate realistic X-ray overlapping. Finally, we train the model with an occlusion curriculum, which enabling multi-granularity localization on X-ray images without manual labels.

the same pixels. *Rather than discovering objects directly from arbitrary cluttered scans, LAO-X constructs self-supervision from mined salient X-ray objects and uses a physics-guided generator to synthesize superposed training samples*.

3 Method

As shown in Fig. 2, LAO-X consists of three main components. First, we develop a **saliency-guided X-ray object mining** module to localize foreground objects from unlabeled scans by contrasting instance-specific regions against trivial backgrounds, thereby deriving an X-ray object bank of object-centric samples. Then, we propose **physics-guided X-ray inspection synthesis and labeling** to synthesize cluttered baggage scans with physics-guided density, which been further annotated with holistic–part pseudo labels. Finally, we introduce a **granularity supervision with occlusion-controllable training** framework to fine-tune a granularity-conditioned model with the synthesized data in an curriculum manner to enable universal object localization in realistic X-ray scenes. Overall, LAO-X is fully automatic and self-supervised, requiring *no* manual annotations.

3.1 Saliency-guided X-ray Object Mining

The variation of objects is the core for training models of universal object localization in X-ray scans. However, standard unsupervised object discovery (*e.g.*, MaskCut [43]) often fails to capture such objectness since faint, low-attenuation items blend into the background, while complex objects with high intensity variations appear fragmented. Thus, we propose Saliency-guided X-ray Object Mining (SXOM) to separate foreground object of various density and structures via a saliency scoring mechanism, and further enhanced by Dual-Gated Saliency Masking (DGSM) (See the top of Fig. 2).

Specifically, let $\mathcal{T} = \{T_n\}_{n=1}^{|\mathcal{T}|}$ be an unlabeled X-ray dataset where each sample $T_n = (\mathbf{H}_n, \mathbf{L}_n, I_n)$ consists of the high-energy density map $\mathbf{H}_n$, the low-energy density map $\mathbf{L}_n$, and a pseudo-RGB image I_n that obtained by mapping densities to color. We first extract patch features $P_n = \{p_n^m\}_{m=1}^{M}$ using a frozen encoder $\phi(\cdot) : \mathbb{R}^{H\times W\times 3} \to \mathbb{R}^{d\times d}$. Then, to quantify the objectness, we define the

saliency score as the average of the smallest $K = \lceil X \cdot (|\mathcal{T}| - 1) \rceil$ distances between patch features:

$$s_n^m = \frac{1}{K} \sum_{i=1}^{K} \left(\text{sort}(d_{n,k}^m)_{k \neq n}\right)_i, \; d_{n,k}^m = \min_{1 \leq j \leq M} \|p_n^m - p_k^j\|_2, \tag{1}$$

where $d_{n,k}^m$ is the the nearest neighbor for patch feature p_n^m in image I_k.

consequently, we upsample these scores into an image-level saliency map $S_n \in [0,1]^{H \times W}$, which reveal the location of foreground objects. However, the raw saliency maps are often coarse due to internal object heterogeneity and density variation. Hereby, we propose a Dual-Gated Saliency Masking (DGSM) module to refine S_n via two sigmoid-based gates with thresholds τ_1, τ_2:

$$G_{\text{sharp}} = \sigma(k \cdot (\tau_1 \max S_n - S_n)), \quad G_{\text{soft}} = \sigma(k \cdot (\tau_2 \max S_n - S_n)). \tag{2}$$

G_{sharp} identifies confident object cores, while G_{soft} captures the broader extent. We fuse these into an adaptive weight map $W_n = \frac{1}{2}(\text{norm}(S_n^3 \odot G_{\text{soft}}) + G_{\text{sharp}})$, where the cubic term S_n^3 suppresses remaining background noise. We then generate a saliency-enhanced image $I_n^{\text{fuse}} = (I_n + C_n) \odot W_n$, making the foreground significantly more distinct for better mining.

With the enhanced X-ray image, we can easily extract a clean instance mask Y_n by adopting any off-the-shelf segmenter $\psi(\cdot)$. Finally, we crop all aligned modalities to form an **X-ray Item Bank** $\mathcal{B} = \{\mathcal{O}_n\}_{n=1}^{|\mathcal{B}|}$, where each entry is a quadruple:

$$\mathcal{O}_n = (\mathbf{H}_n \odot Y_n, \mathbf{L}_n \odot Y_n, I_n \odot Y_n, Y_n) = (\hat{\mathbf{H}}_n, \hat{\mathbf{L}}_n, \hat{I}_n, Y_n). \tag{3}$$

This bank $\mathcal{B}$ serves as the source for the controllable synthesis and pseudo-labeling described in Sec. 3.2, which contains dual-energy density maps $(\hat{\mathbf{H}}, \hat{\mathbf{L}})$ and an isolated object in pseudo-RGB image $\hat{I}$ with the corresponding mask Y

3.2 Physics-guided X-ray Inspection Synthesis and Labeling

Real baggage scans are heavily cluttered: objects overlap extensively, and their appearance changes follow X-ray transmission physics rather than alpha blending. Moreover, manually fully annotate X-ray images is impractical due to the characteristics of X-rays, such as volume superposition, blurred boundaries, and low-attenuation targets. Hence, we propose a Physics-guided X-ray Inspection Synthesis and Labeling (PXISL) process to simulate the real-world X-ray imagery with the X-ray item bank $\mathcal{B}$. PXISL provides physics-guided density superposition and multi-granular holistic–part pseudo labels for both object-level and complementary fine-structure supervision to train LAO-X.

Given the X-ray item bank $\mathcal{B}$, we synthesize a cluttered scan by placing N sampled items $\{\mathcal{O}_i | \mathcal{O}_i \in \mathcal{B}\}_{i=1}^{N}$ onto a real background $(\mathbf{H}^{\text{bg}}, \mathbf{L}^{\text{bg}})$. For each sampled item, we generate a placement transform $\mathcal{J}_i$ with controllable position and random data augmentation. Hence, the canvas-aligned maps is defined as: $(\bar{\mathbf{H}}_i, \bar{\mathbf{L}}_i, \bar{Y}_i) = \mathcal{J}_i(\hat{\mathbf{H}}_{n_i}, \hat{\mathbf{L}}_{n_i}, Y_{n_i})$, where $\bar{Y}_i$ indicates the placed region on the canvas. To simulate the realistic X-ray scans, we project the density into the *absorbance* domain following the Beer–Lambert law [38], *i.e.*, absorbance is additive in log-transmittance for overlap. For each density map $e \in \{H, L\}$, we convert intensities to absorbance:

$$A_e^{\text{bg}} = -\log\left(\frac{e^{\text{bg}}}{\alpha} + \epsilon\right), \quad A_e^{(i)} = -\log\left(\frac{\bar{e}_i}{\alpha} + \epsilon\right), \tag{4}$$

where $\bar{e}_i \in \{\bar{H}_i, \bar{L}_i\}$, α is the incident intensity, and ϵ is a small constant. The overall intensities is calculated by converting the composite absorbance:

$$A_e^{\text{syn}} = A_e^{\text{bg}} + \sum_{i=1}^{K} \left(\bar{Y}_i \odot A_e^{(i)}\right), \qquad e^{\text{syn}} = \alpha \exp(-A_e^{\text{syn}}). \tag{5}$$

Finally, we render the pseudo-RGB visualization as $I^{\text{syn}} = f_{\text{color}}(H^{\text{syn}}, L^{\text{syn}})$, where f_{color} is a standard density to color mapping function. With this controllable synthesis process, we can adjust the complexity of synthesized scans via object number N and the overlap ratio between objects. Furthermore, we obtain the instance-level masks $\{\bar{Y}_i\}_{i=1}^{N}$ under this controllable synthetic process.

Nevertheless, holistic masks alone are not sufficient for X-ray overlaps, which are often **Matryoshka**-like [18]: a visible "part" can itself correspond to an entire object (or a meaningful subcomponent), so

the learner must handle different mask granularities. To obtain object-consistent part masks, we run a segment model $\overline{\psi}(\cdot)$ on the synthesized pseudo-RGB image I^{syn} to produce an over-segmentation $\{R_j\}_{j=1}^{J}$, and filter out masks that deviated from the foreground area:

$$\mathcal{R}^{\text{syn}} = \{\, R_j \mid R_j \subseteq Y^{\text{syn}} \,\}, \quad \text{where } Y^{\text{syn}} = \bigcup_{i=1}^{N} \bar{Y}_i, \tag{6}$$

which preserves local mask with natural visual cues. Then, we design a composite granularity labeling strategy to encourage model learn the relationship between holistic objects and the components. Following UnSAMv2 [45], we define a normalized granularity score for each part mask R_j:

$$g_j = 0.1 + 0.9 \cdot \frac{\sqrt{|R_j|} - s_{\min}}{s_{\max} - s_{\min}}, \quad s_{\max} = \max_{R \in \mathcal{R}^{\text{syn}}} \sqrt{|R|},\ s_{\min} = \min_{R \in \mathcal{R}^{\text{syn}}} \sqrt{|R|}. \tag{7}$$

Then, we further assign $g = 1$ to each holistic instance mask $\bar{Y}_i$, and remove redundant masks with highly overlap. Overall, each synthesized sample is paired with a label pool:

$$\mathcal{G}^{\text{syn}} = \{(\bar{Y}_i, 1)\}_{i=1}^{N} \cup \{(R_j, g_j)\}_{R_j \in \mathcal{R}^{\text{syn}}}. \tag{8}$$

which been further used to train the granularity-conditioned SAM2 in Sec. 3.3.

3.3 Granularity Supervision with Occlusion-Controllable Training

Now we can transfer the "localize any object" ability from RGB foundation models by fine-tuning SAM2 with the controllable synthesized scans and the corresponding holistic–part label pool. Note that, the inherent domain gap between natural RGB and X-ray images comes from the complexity of overlapped structure and low texture visual cues. To address this, we adopt an *occlusion-controlled curriculum* and *granularity-conditioned annotations*, which together stabilize adaptation with object-level and complementary fine-structure supervision from single objects to severe overlap.

Occlusion-controlled curriculum. Our synthesis controls difficulty of curriculum via (i) the number of pasted items N and (ii) an overlap cap γ (maximum overlap between instance masks defined by IoU). We generate tiers $\{\mathcal{D}^{(t)}\}_{t=1}^{T}$ with fixed $(N^{(t)}, \gamma^{(t)})$ and train from easy to hard settings (Model training in Sec. 4.1), progressively increasing clutter and superposition.

Granularity-conditioned SAM2 training. Follow UnSAMv2 [45], we augment SAM2 with a lightweight granularity encoder that injects a continuous $g \in (0, 1]$ into the mask prediction pathway, enabling mask prediction at the requested granularity. We train on synthesized scans paired with our multi-granularity label pool (holistic masks with $g{=}1$ and part masks with $g{<}1$), which teaches SAM2 to localize both whole instances and meaningful regions under *Matryoshka-like* X-ray overlaps.

4 Experiments

4.1 Implementation Details

Physics-guided synthesis: For foreground saliency extraction (Section 3.1), we use a frozen CLIP image encoder [31] with ViT-L/14 [10] as the feature extractor ϕ. For the saliency computation (Eq. (1)), we set $X = 30\%$, and for the dual-gated masking (Eq. (2)), we set $\tau_1 = 0.8$, $\tau_2 = 0.6$, and $k = 5$. Following UnSAMv2 [45], we use CutLER [43] with a Cascade Mask R-CNN [2] object detector as the off-the-shelf segment model. We generate 18,000 synthetic training scenes using the physics-guided Beer–Lambert [38] composition in Section 3.2 by sampling objects uniformly from the mined item bank and applying geometric transforms (*e.g.*, translation, scaling, rotation).

Model training: We adopt a pretrained SAM2-small [32] as the base model for LAO-X and adapt the granularity-conditioned SAM2 design from UnSAMv2 [45] to learn from holistic and part pseudo labels. We fine-tune the fourier encoder, its corresponding granularity mask token, and the transformer block in the mask decoder using LoRA [15] of rank 8 while freezing the rest of the parameters. The batch size is set to 16, and we adopt AdamW [24] with weight decay of 1e-5, and we use cosine scheduler to adjust learning rate during training. All experiments are run on 8 NVIDIA H100 GPUs (80GB). More details of occlusion-controllable curriculum learning can be found in the Appendix D.

Datasets: We use 1621 images in COMPASS-XP [14] as the X-ray scan classification dataset $\mathcal{T}$ for mining objects and synthesizing training data. Following prior X-ray evaluation practice [12],

Table 1: **AP on DET-COMPASS and Five Real-World X-ray Datasets.** The best results are marked in bold, and the second-best results are underlined.

Method	DET-COMPASS			CLCXray			DvXray			PIXray			HiXray			PIDray		
	AP_{50}	AP_{75}	AP_{50}^{95}	AP_{50}	AP_{75}	AP_{50}^{95}	AP_{50}	AP_{75}	AP_{50}^{95}	AP_{50}	AP_{75}	AP_{50}^{95}	AP_{50}	AP_{75}	AP_{50}^{95}	AP_{50}	AP_{75}	AP_{50}^{95}
$RAXO_0^{100}$	54.20	48.80	47.90	29.60	24.40	22.20	26.60	24.10	22.60	45.00	39.00	36.90	27.20	19.40	17.10	21.40	17.90	16.50
$RAXO_{100}^0$	19.80	16.80	16.10	9.70	7.50	7.10	14.10	12.90	12.40	19.80	16.80	16.10	13.00	8.70	7.90	17.10	14.30	13.40
SAM2	36.10	29.13	29.74	17.75	14.73	12.80	10.38	9.72	8.87	6.33	5.45	5.14	18.99	11.22	10.58	1.97	1.16	1.13
SAM3	11.46	9.93	9.69	28.14	21.74	18.93	11.77	9.82	8.55	12.04	9.98	9.05	24.32	12.15	12.77	7.21	3.66	3.79
UnSAMv2	33.25	27.90	27.44	30.71	20.22	19.06	19.58	15.34	14.05	22.54	17.72	16.97	43.16	21.14	22.26	9.57	4.79	4.94
LAO-X	**63.39**	**55.73**	**53.33**	**37.61**	**27.44**	**25.29**	**23.56**	**18.33**	**16.87**	**30.26**	**22.40**	**21.37**	**43.35**	**21.75**	**22.47**	**11.61**	**5.16**	**5.65**

Table 2: **AR on DET-COMPASS and Five Real-World X-ray Datasets.** Measuring item coverage for high confidence predictions.

Method	DET-COMPASS		CLCXray		DvXray		PIXray		HiXray		PIDray	
	AR_1	AR_{10}	AR_1	AR_{10}	AR_1	AR_{10}	AR_1	AR_{10}	AR_1	AR_{10}	AR_1	AR_{10}
SAM2	40.49	47.99	19.52	29.72	19.60	25.66	5.36	21.63	8.91	28.04	5.94	8.63
SAM3	14.11	52.10	23.03	**51.41**	17.82	34.93	5.01	35.95	9.59	**38.92**	8.76	35.68
UnSAMv2	42.88	48.64	21.84	33.73	25.46	33.66	**14.70**	34.83	**17.51**	34.84	**13.62**	23.34
LAO-X	**68.48**	**82.46**	**28.63**	50.20	**30.54**	**46.44**	13.52	**51.86**	16.70	34.97	12.54	**37.06**

Table 3: Segmentation Metrics on PIXray and PIDray.

Method	PIXray		PIDray	
	mIoU	F1	mIoU	F1
SAM2	24.68	28.15	15.15	19.08
SAM3	44.16	49.79	46.48	57.24
UnSAMv2	51.52	60.20	38.24	48.87
LAO-X	**64.06**	**71.86**	**50.37**	**61.17**

we report results on six datasets. DET-COMPASS [12] and five prohibited-item datasets (*i.e.*, CLCXray [46], DvXray [26], PIXray [25], HiXray [39] and PIDray [41]). All six datasets are test-only for LAO-X, as we do not use any of their images for mining, synthesis, or training.

Evaluation Protocol: We evaluate category-agnostic localization with standard MS-COCO box metrics [23]. For SAM-style segmenters, predicted masks are converted to tight bounding boxes. Since several X-ray benchmarks provide only partial annotations, we restrict evaluation to predictions near annotated targets, *i.e.*, only consider predictions which center fall inside all expanded GT boxes by $1.5\times$. COCO matching and aggregation are then applied unchanged, reporting AP_{50}, AP_{75}, mAP, and $AR_{1/10}$. For PIXray and PIDray, which provide pixel-level masks, we also report standard segmentation metrics, mIoU and F1, in Tab. 3. Note that, RAXO [12] results are adopted from the original paper under the conventional MS-COCO evaluation.

4.2 Comparison with the State of the Art

We compare LAO-X with recent X-ray detection and universal segmentation baselines. RAXO [12] is an open-vocabulary X-ray detector that builds dataset-specific class descriptors from either in-house X-ray data ($RAXO_0^{100}$) or Internet data ($RAXO_{100}^0$). Unlike RAXO, LAO-X uses no target category names, text queries, web retrieval, or target-domain annotations. We also evaluate SAM2 [32] and SAM3 [4] as strong RGB promptable segmentation baselines. Finally, UnSAMv2 [45] is fine-tuned on the same COMPASS-XP [14] images but without our object mining, physics-guided synthesis, or occlusion curriculum, isolating the gains from LAO-X's X-ray-specific self-supervised adaptation.

Quantitative results. Tabs. 1 and 2 summarize bbox localization results on DET-COMPASS and five real-world X-ray benchmarks. Overall, LAO-X improves both localization precision and object coverage under limited predictions. On exhaustively annotated DET-COMPASS, LAO-X achieves the best AP_{50}, AP_{75}, and mAP, with {63.39%, 55.73%, 53.33%}, substantially outperforming all baselines. On the five prohibited-item datasets, LAO-X consistently surpasses promptable segmentation baselines in AP_{50}^{95}, *e.g.*increasing UnSAMv2 from 19.06% to 25.29% on CLCXray. It also achieves the best AR_{10} on four of six benchmarks, *e.g.*improving over SAM3 from 34.93% to 46.44% on DvXray. These results show that LAO-X effectively adapts SAM-style localization to X-ray superposition through physics-guided synthesis and occlusion-controlled fine-tuning. RAXO remains a strong semantic detector on prohibited-item benchmarks, while LAO-X surpasses it on DET-COMPASS and achieves the best AP on CLCXray and HiXray, highlighting complementary advantages beyond vocabulary-driven detection.

For PIXray and PIDray, which provide pixel-level masks, we further report standard **segmentation metrics** in Tab. 3. LAO-X achieves the best mIoU and F1 on both datasets, *e.g.*improving over the strongest baseline by more than 10% on PIXray, reaching 64.06%/71.86%. This confirms that LAO-X learns more precise object structure from curriculum training on synthesized scans.

Qualitative comparison. Fig. 3 visualizes the top-5 confident predictions whose box centers fall inside expanded GT regions on real baggage scans. RGB foundation models show a clear X-ray

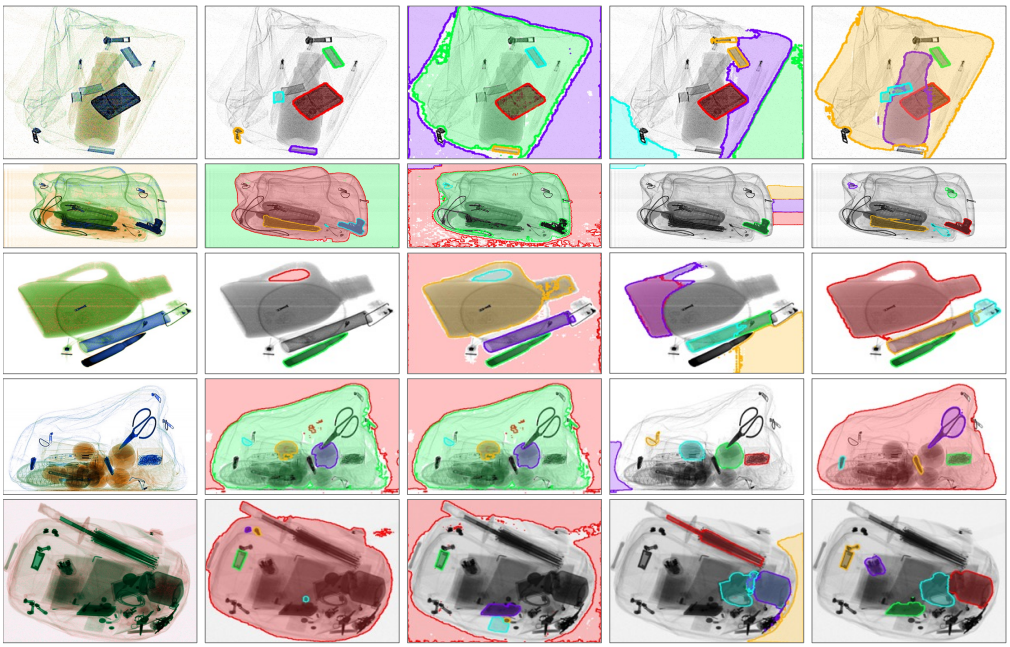


Figure 3: **Comparison of masks with top-5 confidence across real-world X-ray datasets.** To highlight the predicted mask, we convert the original image to a grayscale image in our visualizations. LAO-X produces more potential objects with high confident, *e.g.*, segment the bottle in row 1.

Table 4: Ablation study for our proposed curriculum training strategy for different combinations of synthetic scenes (Easy/Hard/Extreme) and combination of all samples used for curriculum learning.

	Easy	Hard	Extreme	Curriculum	CLCXray					DvXray					PIXray					HiXray					PIDray					Average				
					AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{95}_{50}	AR_1	AR_{10}
(1)	✓	✓	✓		**38.41**	**28.47**	25.21	27.82	40.62	19.99	14.73	13.69	25.63	41.81	24.15	16.57	16.13	9.01	46.77	42.86	22.02	22.33	17.94	33.73	10.28	4.70	5.09	11.68	36.18	27.14	17.30	16.49	18.41	39.82
(2)	✓			✓	35.29	24.84	22.92	26.84	38.92	16.69	12.90	11.66	22.96	35.09	21.17	15.27	14.75	8.00	40.51	**45.07**	**23.42**	**23.65**	**18.46**	**35.11**	8.30	2.90	3.58	9.65	27.38	25.76	16.11	15.58	17.54	34.90
(3)	✓	✓		✓	36.85	27.07	24.16	27.62	36.83	20.86	15.56	14.46	26.89	41.19	25.93	18.48	17.80	10.81	46.70	44.39	22.29	23.08	17.04	34.91	9.58	3.94	4.47	11.09	32.90	27.61	17.48	16.83	18.75	36.45
(4)	✓	✓	✓	✓	37.61	27.44	**25.29**	**28.63**	**50.20**	**23.56**	**18.33**	**16.87**	**30.54**	**46.44**	**30.26**	**22.40**	**21.37**	**13.52**	**51.86**	43.35	21.75	22.47	16.70	34.97	**11.61**	**5.16**	**5.65**	**12.54**	**37.06**	**29.28**	**19.01**	**18.33**	**20.38**	**44.11**

domain gap: SAM2 misses overlapped or low-density objects, such as bottles in the first and third rows, while SAM3 often produces oversized noisy regions spilling into the background. UnSAMv2 improves X-ray-oriented localization but still struggles with heavy superposition, causing unstable extents and incomplete instance separation. In contrast, LAO-X produces more object-aligned high-confidence masks, preserving fine structures and separating overlapping items more consistently. Notably, in the fourth column, LAO-X accurately segments the plastic jug, bullet, and air pump, including the foot peg and valve head. These items are not explicitly boxed by manual annotations, suggesting stronger class-agnostic localization beyond labeled targets.

4.3 Ablation Study

Occlusion-controllable curriculum: To isolate occlusion-controllable training, we compare LAO-X trained on all synthetic scenes at once with the stage-wise Easy $\rightarrow$ Hard $\rightarrow$ Extreme schedule. Tab. 4 shows that curriculum training improves localization across all metrics. Comparing Rows (1) and (4), LAO-X raises mAP from 16.49% to 18.33% over direct training on mixed synthetic samples, indicating that mixed-complexity scenes are less effective for adapting RGB-pretrained SAM-style models to X-rays. Rows (2)–(4) further show consistent gains as occlusion increases gradually, with mAP rising from 15.58% to 16.83% and finally to 18.33%. This confirms that the object number N and occlusion level γ serve as effective curriculum-difficulty measures. Overall, progressively increasing occlusion provides a smoother adaptation path, stabilizing optimization from isolated objects to densely overlapped baggage scans and improving recall under X-ray superposition.

Interactive segmentation comparison: We study the capabilities of LAO-X for interactive segmentation by querying images with the same point prompt yet different granularity values g. As shown in Fig. 4, UnSAMv2 exhibits brittle granularity behavior: changing g can trigger abrupt and often incorrect mask shifts (*e.g.*, expanding to background/baggage regions) or fail to produce meaningful

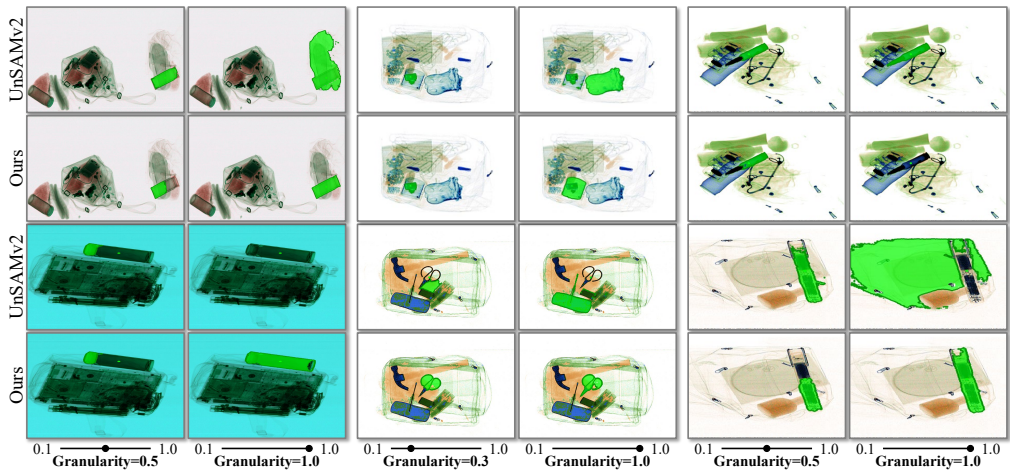


Figure 4: **Granularity comparison with UnSAMv2.** Green overlays are predicted masks at the queried granularity g (shown under each column; same image prompt, only g changes). UnSAMv2 is unstable and may collapse to over-segmenting large regions, while LAO-X follows g more consistently, transitioning from part-level (smaller g) to holistic masks (larger g) under heavy overlap.

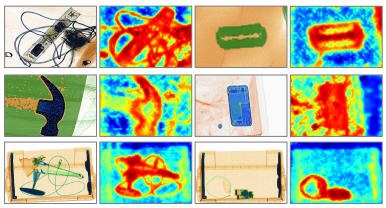


Figure 5: Visualization of Saliency Response.

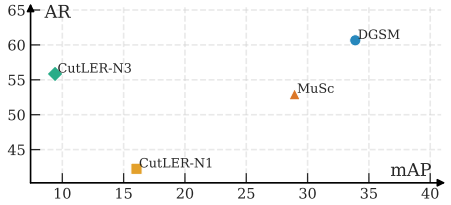


Figure 6: Object bank construction comparison.

part-level masks under overlap. In contrast, LAO-X responds to g in a stable and monotonic manner: smaller g yields compact part-like regions, while a larger g progressively expands toward holistic object masks, all while preserving plausible boundaries and instance separation in cluttered scans. This qualitative study highlights the benefit of our holistic-part supervision and occlusion-controllable curriculum for learning reliable granularity-conditioned localization in X-ray.

Additional analysis of saliency mining. Fig. 5 visualizes the saliency response across various scenes, showing that our scoring mechanism can highlight foreground items with diverse densities, especially for weak attenuation, and complex internal structures. Although such responses are effective across different X-ray scenes, we construct the object bank from COMPASS-XP as it contains richer object categories together with dual-energy scans, offering more diverse density patterns for physics-guided synthesis. Fig. 6 further compares object mining quality on DET-COMPASS. Except for CutLER-N3, which predicts 3 masks, each method produces one mask per image. LAO-X achieves the best mAP and AR, outperforming direct CutLER and MuSc, which indicates that DGSM improves the accuracy and completeness of mined object masks, providing a cleaner object bank for subsequent synthesis.

5 Conclusion

In this paper, we present LAO-X, a self-supervised framework for *Localizing Any Object* in X-ray security scans without manual annotations. LAO-X first mines object-centric items from unlabeled scans via saliency-guided object extraction, recomposes them with physic-guided synthesis to trains a granularity-conditioned SAM2 with an occlusion-controllable curriculum. This pipeline turns unlabeled X-ray data into structured supervision and enables reliable localization of both whole and parts of objects under severe superposition. Experiments on six X-ray benchmarks show consistent improvements over RGB foundation models and self-supervised baselines in precision, recall, and mask quality. One limitation is the domain gap caused by pseudo-color mapping of different scanner, where we can learn a color-mapping sampling model in the future work to enhance synthesis process to improve robustness across scanners. We hope LAO-X can serve as a strong and practical baseline for universal X-ray perception and stimulate future work on scalable X-ray localization.

References

[1] E. Benedykciuk, M. Denkowski, and K. Dmitruk. Material classification in x-ray images based on multi-scale cnn. *Signal, Image and Video Processing*, 15(6):1285–1293, 2021.

[2] Z. Cai and N. Vasconcelos. Cascade r-cnn: High quality object detection and instance segmentation. *IEEE TPAMI*, 43(5):1483–1498, 2019.

[3] S. Cao, J. Gu, J. Kuen, H. Tan, R. Zhang, H. Zhao, A. Nenkova, L. Gui, T. Sun, and Y. Wang. SOHES: self-supervised open-world hierarchical entity segmentation. In *ICLR*, 2024.

[4] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, et al. Sam 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.

[5] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin. Emerging properties in self-supervised vision transformers. In *ICCV*, pages 9650–9660, 2021.

[6] A. Chang, Y. Zhang, S. Zhang, L. Zhong, and L. Zhang. Detecting prohibited objects with physical size constraint from cluttered x-ray baggage images. *Knowledge-Based Systems*, 237: 107916, 2022.

[7] K. Chen, C. Liu, H. Chen, H. Zhang, W. Li, Z. Zou, and Z. Shi. Rsprompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model. *IEEE Transactions on Geoscience and Remote Sensing*, 2024.

[8] T. Chen, L. Zhu, C. Deng, R. Cao, Y. Wang, S. Zhang, Z. Li, L. Sun, Y. Zang, and P. Mao. Sam-adapter: Adapting segment anything in underperformed scenes. In *ICCV*, pages 3367–3375, 2023.

[9] Q. Dai and S. Yang. Curriculum point prompting for weakly-supervised referring image segmentation. In *CVPR*, pages 13711–13722, 2024.

[10] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

[11] M. Fey. *3D printing and international security: risks and challenges of an emerging technology*, volume 144. DEU, 2017.

[12] P. Garcia-Fernandez, L. Vaquero, M. Liu, F. Xue, D. Cores, N. Sebe, M. Mucientes, and E. Ricci. Superpowering open-vocabulary object detectors for x-ray vision. In *ICCV*, pages 20770–20779, 2025.

[13] Y. F. A. Gaus, N. Bhowmik, B. K. Isaac-Medina, and T. P. Breckon. Performance evaluation of segment anything model with variational prompting for application to non-visible spectrum imagery. In *CVPR*, pages 3142–3152, 2024.

[14] L. Griffin, M. Caldwell, and J. Andrews. Compass-xp dataset. *Computational Security Science Group, UCL*, pages 4700–4708, 2019.

[15] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.

[16] S. U. Khan, I. U. Khan, I. Ullah, N. Saif, and I. Ullah. A review of airport dual energy x-ray baggage inspection techniques: Image enhancement and noise reduction. *Journal of X-ray Science and Technology*, 28(3):481–505, 2020.

[17] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, et al. Segment anything. In *ICCV*, pages 4015–4026, 2023.

[18] A. Kusupati, G. Bhatt, A. Rege, M. Wallingford, A. Sinha, V. Ramanujan, W. Howard-Snyder, K. Chen, S. Kakade, P. Jain, et al. Matryoshka representation learning. *Advances in Neural Information Processing Systems*, 35:30233–30249, 2022.

[19] F. Li, H. Zhang, P. Sun, X. Zou, S. Liu, C. Li, J. Yang, L. Zhang, and J. Gao. Segment and recognize anything at any granularity. In *ECCV*, pages 467–484. Springer, 2024.

[20] M. Li, T. Jia, H. Wang, B. Ma, H. Lu, S. Lin, D. Cai, and D. Chen. Ao-detr: Anti-overlapping detr for x-ray prohibited items detection. *IEEE Transactions on Neural Networks and Learning Systems*, 36(7):12076–12090, 2024.

[21] Y. Li, M. Hu, and X. Yang. Polyp-sam: Transfer sam for polyp segmentation. In *Medical Imaging 2024: Computer-Aided Diagnosis*, volume 12927, pages 759–765. SPIE, 2024.

[22] S. Lin, T. Jia, H. Wang, B. Ma, M. Li, and D. Chen. Detection of novel prohibited item categories for real-world security inspection. *Eng. Appl. Artif. Intell.*, 2025.

[23] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. Microsoft coco: Common objects in context. In *ECCV*, pages 740–755. Springer, 2014.

[24] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[25] B. Ma, T. Jia, M. Su, X. Jia, D. Chen, and Y. Zhang. Automated segmentation of prohibited items in x-ray baggage images using dense de-overlap attention snake. *IEEE TMM*, 25:4374–4386, 2022.

[26] B. Ma, T. Jia, M. Li, S. Wu, H. Wang, and D. Chen. Toward dual-view x-ray baggage inspection: A large-scale benchmark and adaptive hierarchical cross refinement for prohibited item discovery. *IEEE Transactions on Information Forensics and Security*, 19:3866–3878, 2024.

[27] B. Ma, T. Jia, M. Li, S. Wu, H. Wang, and D. Chen. Toward dual-view x-ray baggage inspection: A large-scale benchmark and adaptive hierarchical cross refinement for prohibited item discovery. *IEEE Transactions on Information Forensics and Security*, 19:3866–3878, 2024.

[28] R. D. Macdonald. Design and implementation of a dual-energy x-ray imaging system for organic material detection in an airport security application. In *Machine Vision Applications in Industrial Inspection IX*, volume 4301, pages 31–41. SPIE, 2001.

[29] Y. Michael, M. Alansari, A. Ahmed, N. Werghi, and A. Henschel. X-ssl: Self-supervised x-ray threat detection with zero-shot and multi-modal learning. *Information Processing & Management*, 63(1):104339, 2026.

[30] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

[31] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763. PmLR, 2021.

[32] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.

[33] V. Rebuffel and J.-M. Dinten. Dual-energy x-ray imaging: benefits and limits. *Insight-non-destructive testing and condition monitoring*, 49(10):589–594, 2007.

[34] J. Shi and J. Malik. Normalized cuts and image segmentation. *IEEE TPAMI*, 22(8):888–905, 2000.

[35] H. Sima, B. Chen, C. Tang, Y. Zhang, and J. Sun. Multi-scale feature attention-detection transformer: Multi-scale feature attention for security check object detection. *IET Computer Vision*, 2024.

[36] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa, et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025.

[37] A. Singh and Dhiraj. Advancements in machine learning techniques for threat item detection in x-ray images: a comprehensive survey. *Int. J. Multim. Inf. Retr.*, 2024.

[38] D. F. Swinehart. The beer-lambert law. *Journal of chemical education*, 39(7):333, 1962.

[39] R. Tao, Y. Wei, X. Jiang, H. Li, H. Qin, J. Wang, Y. Ma, L. Zhang, and X. Liu. Towards real-world x-ray security inspection: A high-quality benchmark and lateral inhibition module for prohibited items detection. In *ICCV*, pages 10923–10932, 2021.

[40] R. Tao, H. Li, T. Wang, Y. Wei, Y. Ding, B. Jin, H. Zhi, X. Liu, and A. Liu. Exploring endogenous shift for cross-domain detection: A large-scale benchmark and perturbation suppression network. In *CVPR*, 2022.

[41] B. Wang, L. Zhang, L. Wen, X. Liu, and Y. Wu. Towards real-world prohibited item detection: A large-scale x-ray benchmark. In *ICCV*, pages 5412–5421, 2021.

[42] R. Wang, Y. Shi, and M. Cai. Optimization and research of suspicious object detection algorithm in x-ray image. In *2023 IEEE 6th Information Technology,Networking,Electronic and Automation Control Conference (ITNEC)*, 2023.

[43] X. Wang, R. Girdhar, S. X. Yu, and I. Misra. Cut and learn for unsupervised object detection and instance segmentation. In *CVPR*, pages 3124–3134, 2023.

[44] X. Wang, J. Yang, and T. Darrell. Segment anything without supervision. *NeurIPS*, 37: 138731–138755, 2024.

[45] J. Yu, T. Darrell, and X. Wang. Unsamv2: Self-supervised learning enables segment anything at any granularity. *arXiv preprint arXiv:2511.13714*, 2025.

[46] C. Zhao, L. Zhu, S. Dou, W. Deng, and L. Wang. Detecting overlapped objects in x-ray security imagery by a label-aware mechanism. *IEEE transactions on information forensics and security*, 17:998–1009, 2022.

[47] Y. Zhao, K. Li, Z. Cheng, P. Qiao, X. Zheng, R. Ji, C. Liu, L. Yuan, and J. Chen. Graco: Granularity-controllable interactive segmentation. In *CVPR*, pages 3501–3510, 2024.

Overview

In this appendix, we provide additional descriptions of the following contents:

- Implementation Details (Section A).
- Segmentation evaluation on real X-ray inspection dataset (Section B).
- Pseudo label of holistic/partial objects and their combination (Section C).
- Complexity of synthetic instances via objects number (K) and overlap range (γ) (Section D).
- Qualitative comparison of mask localization performance (Section E).
- Quantitative and Qualitative analysis of interactive mask prediction across multi-granularity (Section F).
- Broader Impact discussion (Section G).

Table 5: Segmentation evaluation results on PIXray and PIDray (using standard COCO metric).

Method	**PIXray**					**PIDray**				
	AR_1	AR_{10}	AP_{50}	AP_{75}	AP_{50}^{95}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP_{50}^{95}
SAM2	0.64	20.42	3.48	3.22	2.85	3.18	8.19	0.89	0.53	0.51
SAM3	0.60	29.74	6.54	5.41	4.88	2.24	2.16	2.57	1.04	1.19
UnSAMv2	**10.70**	<u>29.20</u>	<u>13.60</u>	<u>11.02</u>	<u>10.41</u>	**5.81**	<u>15.61</u>	**3.80**	<u>1.59</u>	<u>1.83</u>
LAO-X	6.85	**41.94**	**15.29**	**11.80**	**10.93**	<u>4.35</u>	**24.14**	<u>3.79</u>	**1.78**	**1.88**

A More on Implementation Details

Implementation Details of Promptable Segmentation Baselines. We evaluate SAM2 [32] and SAM3 [4] in the automatic mask generation setting for zero-shot X-ray promptable instance segmentation. Meanwhile, wecalibrating their mask thresholds on each dataset's training split and reporting the best validated results for fairness. No ground-truth boxes, points, masks, or category texts are used as prompts. Instead, each model internally samples a regular grid of point prompts over the input image and generates a set of candidate masks, which are then filtered by the model's predicted mask quality, stability score, binarization threshold, and non-maximum suppression.

For SAM2, we use the SAM2.1 small checkpoint with the Ultralytics SAM interface. Images are resized to a square resolution of 1024×1024, and automatic mask generation uses the default point grid. For SAM3, we use the official image model with the HuggingFace mask-generation pipeline. Images are processed in RGB format, and the automatic mask generation pipeline samples a denser point grid. We use point-prompt automatic generation only, without text prompts, ground-truth annotations, or multi-scale crop inference.

Since mask binarization and confidence filtering can strongly affect automatic mask generation results, we calibrate the relevant thresholds on the validation split of each dataset while keeping the remaining generation settings fixed. All predictions are saved in COCO instance segmentation format, and predicted masks are converted to tight bounding boxes for box-level evaluation. For class-agnostic evaluation, all predicted instances are assigned to a single foreground category.

Dual-energy X-ray imaging. X-ray security scanners can be implemented with different acquisition modes, each providing a distinct trade-off between deployability, geometric coverage, and material awareness. Single-energy transmission imaging records only line-integrated attenuation, and is therefore simple and widely used, but it entangles material thickness, density, and composition in a single response. Dual-energy X-ray imaging mitigates this ambiguity by acquiring low-energy and high-energy measurements, whose energy-dependent attenuation differences provide material-related cues such as effective atomic number and density, and are commonly used to render pseudo-color maps for security screening [33, 28, 16, 1].

In contrast, dual-view or multi-view systems improve geometric coverage by imaging baggage from multiple projection angles, but do not by themselves provide material decomposition unless combined with spectral sensing. CT-based EDS further reconstructs volumetric structure and can provide stronger three-dimensional analysis, at the cost of more specialized hardware and deployment

Table 6: Ablation study of pseudo-labeling granularity. We evaluate the performance of holistic instance masks, over-segmented local parts, and the proposed unified LAO-X strategy across five X-ray benchmarks.

Labeling Strategy	CLCXray			DvXray			PIXray			HiXray			PIDray		
	AP_{50}	AP_{75}	AP^{95}_{50}	AP_{50}	AP_{75}	AP^{95}_{50}	AP_{50}	AP_{75}	AP^{95}_{50}	AP_{50}	AP_{75}	AP^{95}_{50}	AP_{50}	AP_{75}	AP^{95}_{50}
Standard COCO evaluation															
Holistic Object	17.46	13.65	12.24	3.97	2.91	3.16	15.04	11.48	11.04	16.73	8.74	8.83	**6.49**	**2.98**	2.72
Partial Object	18.72	15.57	13.23	**5.05**	**4.59**	**4.02**	8.81	6.41	6.52	26.03	13.57	13.66	5.89	2.40	**2.80**
LAO-X	**23.90**	**18.40**	**16.16**	4.81	4.32	3.77	**16.03**	**11.93**	**11.38**	**26.30**	**13.67**	**13.80**	5.42	2.48	2.68

constraints, while backscatter imaging offers complementary surface-sensitive contrast but is not a replacement for transmission-based high-energy and low-energy measurements. Thus, dual-energy channels provide valuable physics-grounded material cues when available, but their accessibility is scanner-dependent rather than universal. In our PXISL module, using dual-energy maps allows Beer–Lambert composition to be performed independently in the high- and low-energy absorbance domains, preserving material-dependent attenuation during synthesis. When only single-energy X-ray scans are available, Eqs. 4 and 5 naturally reduce to a single-channel density-map-based synthesis by applying the same absorbance-domain addition to the available attenuation map.

B Segmentation evaluation on real X-ray inspection dataset

We select PIXray and PIDray to evaluate segmentation since they provide pixel-level masks for hazardous objects. Tab. 5 reports COCO-style mask metrics. Across both datasets, LAO-X delivers the best overall segmentation quality: it achieves the highest mIoU / F1 on PIXray (64.06% / 71.86%) and PIDray (50.37% / 61.17%), outperforming UnSAMv2 by large margins. COCO metrics further show that LAO-X improves object coverage under clutter, reaching the best AR_{10} on both PIXray (41.94% vs. 29.20%) and PIDray (24.14% vs. 15.61%). On PIXray, LAO-X also yields the highest precision (AP_{50}: 15.29%, AP^{95}_{50}: 10.93%). On PIDray, LAO-X remains comparable on AP_{50} (3.79% vs. 3.80%) while improving stricter metrics (AP_{75}: 1.78% vs. 1.59%; AP^{95}_{50}: 1.88% vs. 1.83%), indicating more accurate masks when localized. Overall, the consistent gains in both region-based and COCO-style metrics confirm that LAO-X produces cleaner masks with higher coverage on real-world X-ray scans.

C Pseudo label of holistic/partial objects and their combination

To evaluate the impact of multi-granularity supervision, we compare training with holistic masks derived from saliency-guided mining (Holistic), over-segmented local components (Part), and our combined strategy (LAO-X). As shown in Tab. 6, LAO-X significantly outperforms holistic or part pseudo-labels across most benchmarks. For instance, on CLCXray, LAO-X achieves 23.90% AP_{50}, markedly higher than Holistic (17.46%) or Part (18.72%) alone. This improvement is even more evident in HiXray, where AP_{50} decreases from 16.73% to 26.30% (+9.57). These results demonstrate that holistic and part-level annotations provide complementary information: while part labels handle internal complexity, for example in DvXray (5.05% AP_{50}), our integrated approach ensures the most robust and precise localization across various distribution of X-ray targets. By effectively merging these multi-granularity cues, LAO-X learns the complex, multi-overlapped structures inherent in security scans.

D Complexity of synthetic instances via objects number and overlap range

Our occlusion-controlled curriculum learning strategy (Section 3.3) consists of Easy / Hard / Extreme stages, increasing both the number of pasted items N of $\{[2,4]\ /\ [4,7]\ /\ [6,9]\}$ and the overlap level γ of $\{[0.0,0.05]\ /\ [0.0,0.3]\ /\ [0.0,0.5]\}$, where the learning rate and training epochs is set for $\{1.6\times10^{-4}\ /\ 1.2\times10^{-4}\ /\ 1.0\times10^{-4}\}$, and $\{80\ /\ 64\ /\ 48\}$. We train sequentially by stages and reset the optimizer at the start of each stage.

Moreover, we evaluate framework sensitivity to synthesis complexity by varying the object count (K) and overlap ratio (γ) during the extreme stage. As illustrated in Tab. 7, performance remains

Table 7: Sensitivity analysis of synthesis parameters. The table reports performance variations across different object counts (K) and overlap ratios (γ) within the extreme curriculum stage.

Synthetic parameter		CLCXray					DvXray					PIXray					HiXray					PIDray					Average				
Objects (K)	Overlap (γ)	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}	AP_{50}	AP_{75}	AP^{m}_{50}	AR_1	AR_{10}
(LAO-X) 6-9	[0.0,0.5]	37.61	27.44	25.29	28.63	50.20	23.56	18.33	16.87	30.54	46.44	**30.26**	**22.40**	**21.37**	13.52	**51.86**	43.35	21.75	22.47	16.70	34.97	**11.61**	5.16	5.65	12.54	37.06	29.28	19.01	18.33	20.38	44.11
9-12	[0.0,0.5]	37.38	27.42	25.17	27.83	49.78	23.89	18.79	17.26	31.11	46.79	29.79	21.93	20.97	13.09	51.60	**44.57**	22.30	**23.10**	**17.00**	**35.64**	11.29	5.20	5.62	12.52	37.10	29.38	19.13	18.42	20.31	44.18
12-15	[0.0,0.5]	**39.01**	**29.30**	**26.46**	**28.77**	**51.77**	**24.25**	**18.94**	**17.52**	**32.00**	**47.99**	30.13	22.16	21.20	**13.81**	51.81	43.86	21.49	22.63	16.78	35.11	11.33	5.37	5.67	**12.89**	37.46	**29.71**	**19.45**	**18.70**	**20.85**	**44.83**
6-9	[0.0,0.7]	36.81	27.06	24.66	27.66	49.33	23.16	18.01	16.56	29.95	45.28	29.57	21.96	20.90	13.43	50.78	44.40	**22.31**	23.06	16.93	35.60	10.76	4.89	5.30	12.17	36.37	28.94	18.85	18.10	20.03	43.47
6-9	[0.0,0.9]	36.22	26.72	24.41	27.04	48.57	23.19	17.97	16.59	30.14	45.05	29.22	21.36	20.42	12.30	50.93	44.10	21.91	22.79	16.87	35.27	11.39	**5.45**	**5.78**	12.76	**38.05**	29.30	18.83	18.07	20.14	42.13

remarkably stable across these configurations, with AP metrics fluctuating only marginally as K increases up to 15 or γ expands to 0.9. Notably, increasing K and γ naturally provides a higher volume of both holistic and part annotations for training. This indicates that leveraging a denser set of supervision signals has the potential to further improve model performance. However, comparing these results with the ablation of curriculum tiers in Table 4 (Main paper) of curriculum learning reveals that the staged design of the curriculum is a much more significant factor than specific stage hyper-parameters. This strengthen our conclusion that the sequential transition from simple to heavily cluttered scenes is the primary driver for optimization stability and generalization in the projective X-ray domain.

E Qualitative Comparison of Mask Localization Performance.

Section D provide additional top-5 confident mask predictions across real-world X-ray datasets, where the original scans are converted to grayscale to make the predicted masks easier to inspect. SAM2 often produces sparse or incomplete masks, especially for low-contrast and overlapped objects, while SAM3 tends to generate larger but noisy regions that frequently spill into background or merge nearby structures. UnSAMv2 improves the objectness of predicted regions, but its mask extents remain unstable under heavy superposition and complex baggage layouts. In contrast, LAO-X produces more object-aligned top-confidence masks, with cleaner boundaries, better instance separation, and stronger preservation of fine structures. These visualizations also show that LAO-X can recover plausible objects beyond the manually annotated prohibited items, supporting its ability to learn a more general X-ray localization prior rather than only fitting labeled target categories.



Figure 7: **More on comparison of masks with top-5 confidence across real-world X-ray datasets.** To highlight the predicted mask, we convert the original image to a grayscale image in our visualizations.



Figure 8: **More on comparison of masks with top-5 confidence across real-world X-ray datasets.** To highlight the predicted mask, we convert the original image to a grayscale image in our visualizations.

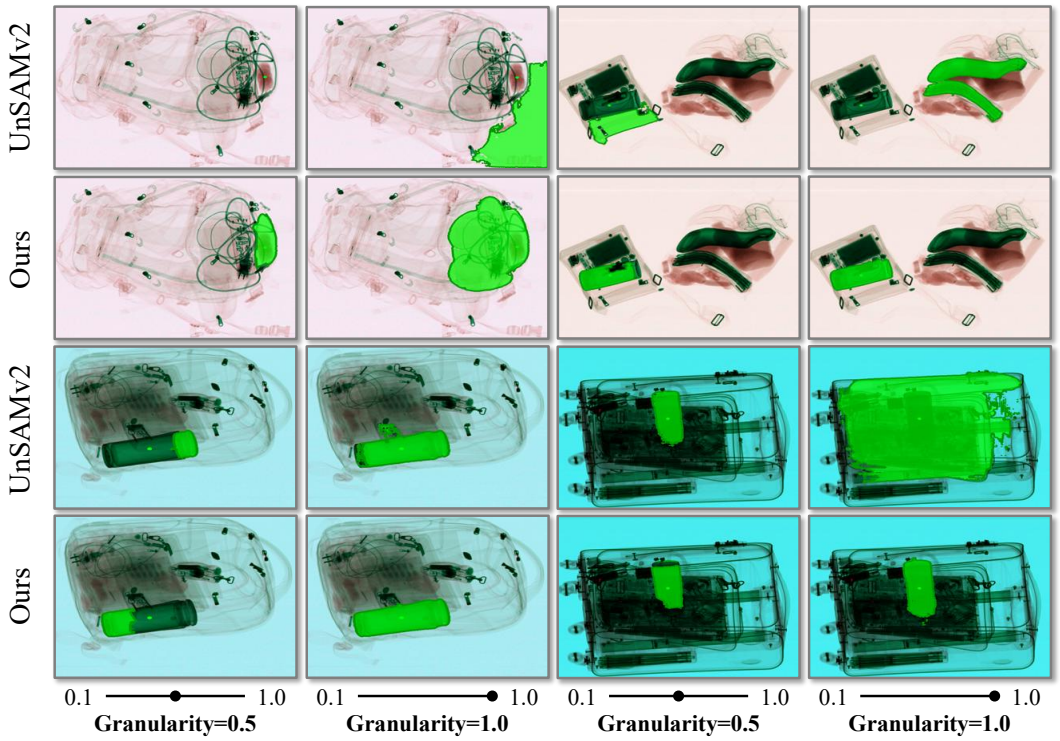


Figure 9: **Granularity comparison with UnSAMv2 on CLCXray.** Green overlays are predicted masks at the queried granularity g (shown under each column; same image prompt, only g changes). UnSAMv2 is unstable and may collapse to over-segmenting large regions, while LAO-X follows g more consistently, transitioning from part-level (smaller g) to holistic masks (larger g) under heavy overlap.

F Quantitatively and Qualitative Analysis of Interactive Multi-Granularity Segmentation.

Quantitatively analysis: Fig. 10 analyzes how the queried granularity $g \in [0.1, 1.0]$ affects localization accuracy of LAO-X. On DET-COMPASS, AP_{50} increases almost monotonically with g, indicating that larger granularity encourages holistic masks that better align with fully annotated boxes. On real-world prohibited-item benchmarks, the optimal g is dataset-dependent: PIDray favor coarse predictions (best around $g \approx 0.8$), while PIXray peaks at mid-level granularity where $g \approx 0.3$, suggesting that overly coarse masks may merge adjacent items under clutter. Though the inherent distribution of object scale shifts across different datasets, the proposed methods maintain strong capability at granularity 1.0. Overall, the sweep confirms that granularity is a practical control knob: LAO-X can adapt its output scale to different datasets and labeling regimes, and we report the optimal g for each benchmark in all quantitative comparisons.

Qualitative analysis: We evaluate the interactive segmentation capabilities of LAO-X by querying images with the same point prompt across different granularity values g. As shown in Figs. 9, 11 and 12, UnSAMv2 exhibits unstable behavior where changing g often triggers incorrect mask shifts, such as expanding into the background or failing to define distinct parts under overlap. In contrast, LAO-X responds to g in a stable and monotonic manner: a smaller g produces compact part-level regions, while a larger g progressively expands toward the holistic object mask. This consistency, even in cluttered scans, confirms that our multi-granularity supervision enables the model to learn reliable, scale-conditioned localization priors for X-ray security imagery.

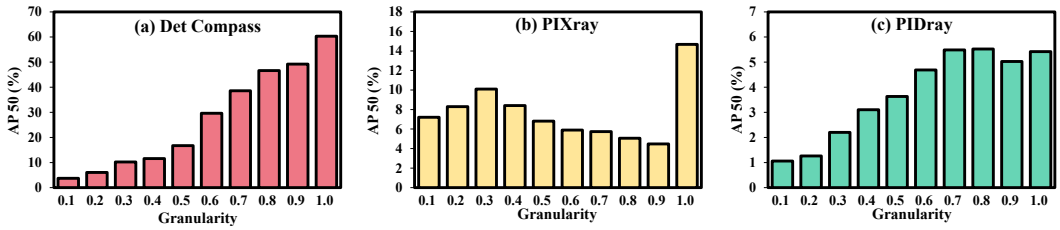
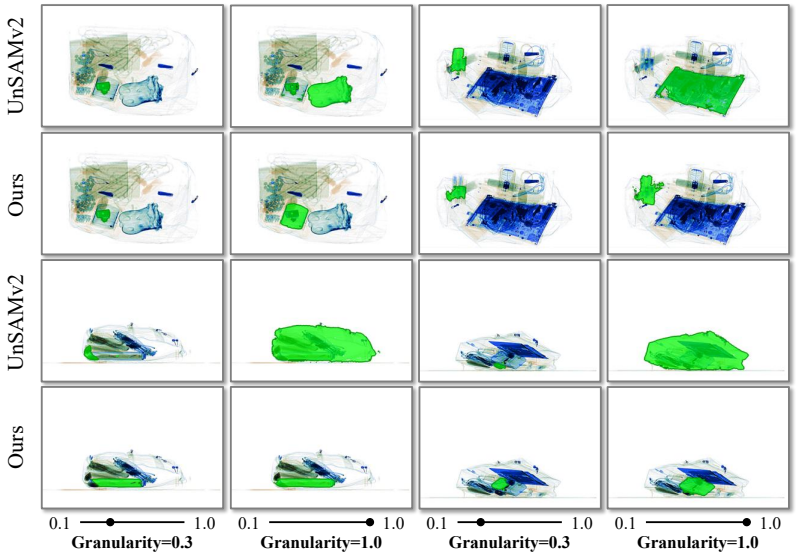


Figure 10: **Granularity sweep of LAO-X.** AP_{50} as a function of the queried granularity $g \in [0.1, 1.0]$ on DET-COMPASS and real-world X-ray benchmarks.

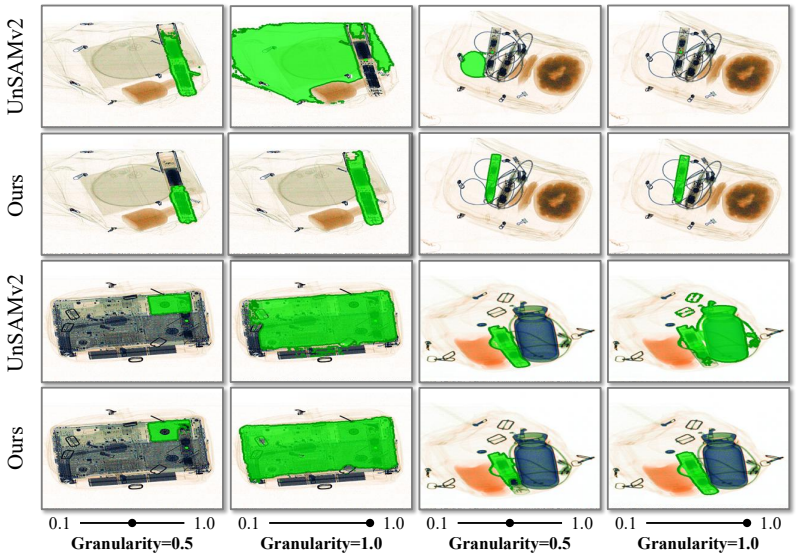
G Broader Impact

The primary motivation of this work is to improve open-world object localization in X-ray security screening. By adapting a promptable localization model to X-ray scans without manual annotations, LAO-X will support beneficial applications such as assisting human operators, improving inspection efficiency, and helping identify unusual or previously unseen objects in cluttered baggage. Because LAO-X learns from unlabeled X-ray data and physics-guided synthetic samples, it may reduce the need for expensive expert annotation and make X-ray perception research more scalable.

At the same time, X-ray security screening is a safety-critical application. Incorrect predictions may have practical consequences: false negatives may miss dangerous items, while false positives may increase unnecessary manual inspection or delay passengers. The method may also be affected by scanner differences, dataset bias, unusual object configurations, or changes in operating conditions. For these reasons, LAO-X should be used as an assistive tool rather than as the sole basis for security decisions. Responsible deployment requires human oversight, careful validation on the target scanner and environment, clear communication of system uncertainty, and continuous monitoring for failure cases after deployment.

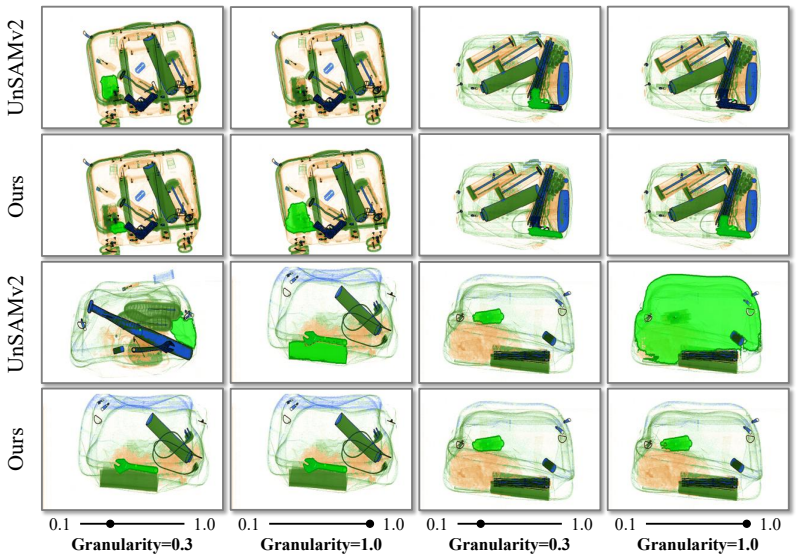


(a) DvXray

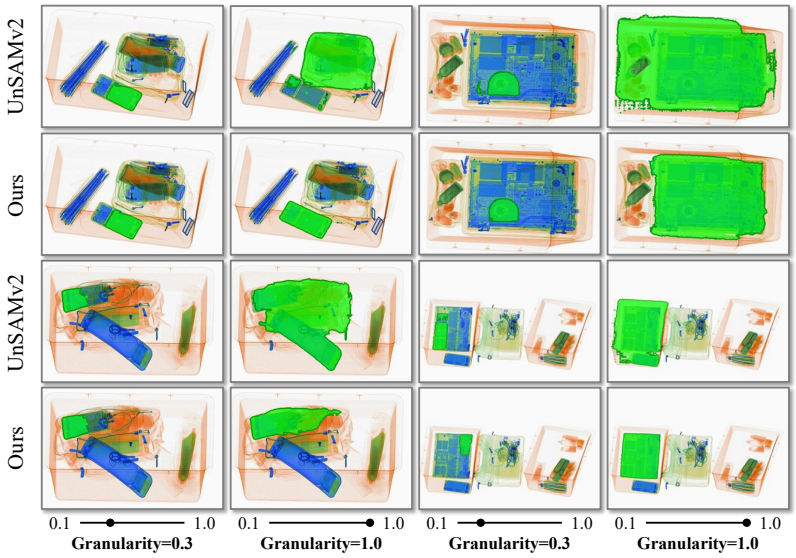


(b) HiXray

Figure 11: More on **Granularity comparison with UnSAMv2 on DvXray and HiXray.**



(a) PIXray



(b) PIDray

Figure 12: More on **Granularity comparison with UnSAMv2 on PIXray and PIDray.**

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim a self-supervised framework for category-agnostic X-ray object localization without human annotations. These claims are supported by the method in Sec. 3, the implementation and evaluation protocol in Sec. 4, and the quantitative and qualitative results on six X-ray benchmarks.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The conclusion discusses limitations related to scanner-dependent pseudo-color mappings and future work on color-mapping sampling for more robust synthesis. Appendix A also discusses scanner-dependent availability of dual-energy channels and how the synthesis reduces to single-channel absorbance-domain composition when only single-energy scans are available.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper does not present formal theoretical results, theorems, or proofs. The mathematical expressions describe the proposed saliency computation, Beer–Lambert-based synthesis, and granularity labeling used in the empirical method.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The paper specifies the training data source, evaluation datasets, saliency-mining procedure, synthesis equations, curriculum stages, model backbone, trainable modules, optimizer, learning rates, batch size, and evaluation protocol in Sec. 3, Sec. 4, and Appendix A. Baseline implementation details for SAM2 and SAM3 are further provided in Appendix A.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
     (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
     (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
     (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Source code and execution instructions will be made publicly available upon publication. The experiments use publicly available datasets and models where applicable, and the paper provides sufficient methodological and experimental details to support reproduction.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The implementation details specify the unlabeled training source, test-only datasets, frozen CLIP encoder, CutLER configuration, number of synthetic scenes, SAM2-small backbone, LoRA rank, optimizer, weight decay, batch size, scheduler, learning rates, epochs, object-count ranges, overlap ranges, and evaluation metrics. Additional segmentation and baseline details are given in the appendix.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: In line with prior X-ray detection and segmentation works, and due to the computational cost of the full training and evaluation pipeline, we report single-run results without error bars or confidence intervals.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification:The paper reports that experiments were run on 8 NVIDIA H100 GPUs with 80GB memory. The total compute cost is estimated to be approximately 50 GPU hours.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: We have reviewed the NeurIPS Code of Ethics and believe that this work conforms to it. The work uses existing X-ray datasets, focuses on assistive security-screening research, and discusses safety-critical deployment considerations.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The paper discusses broader impacts in Appendix G. It describes potential benefits for assisting X-ray security screening and reducing expert annotation needs, while also noting safety-critical risks such as false negatives, false positives, scanner-domain shifts, and the need for human oversight and target-environment validation.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper does not release a new dataset or a high-risk pretrained model. The released artifact is source code for research reproduction, while all component models and datasets remain subject to their original usage terms.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer:[Yes]

Justification: All models and datasets used in this work, including COMPASS-XP, DET-COMPASS, CLCXray, DvXray, PIXray, HiXray, PIDray, CLIP, SAM2, SAM3, CutLER, UnSAMv2, and LoRA, are cited in the manuscript. The released code will document the corresponding licenses and usage terms of these assets.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The only new asset released is the LAO-X source code, which will be made publicly available upon publication. Sufficient methodological detail to reimplement the method independently is provided in Section 3 and Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing experiments or research with human subjects. It uses existing X-ray datasets and self-supervised training from unlabeled scans.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve human subjects, crowdsourcing, or participant studies, so IRB approval is not applicable.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The core method does not use LLMs as a methodological component. Any use of LLMs is solely for grammar and formatting assistance.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.